2 **(a)** Describe an experiment to determine the spring constant of a spring.

State:

- the apparatus you need
- details of how to take measurements
- how to calculate the spring constant

You may use the space below to draw a **labelled** diagram as part of your answer.

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(b) Fig. 2.1 shows a baby in a baby bouncer. The baby bouncer consists of a holder suspended from a spring. The baby pushes his feet on the ground and bounces gently up and down.

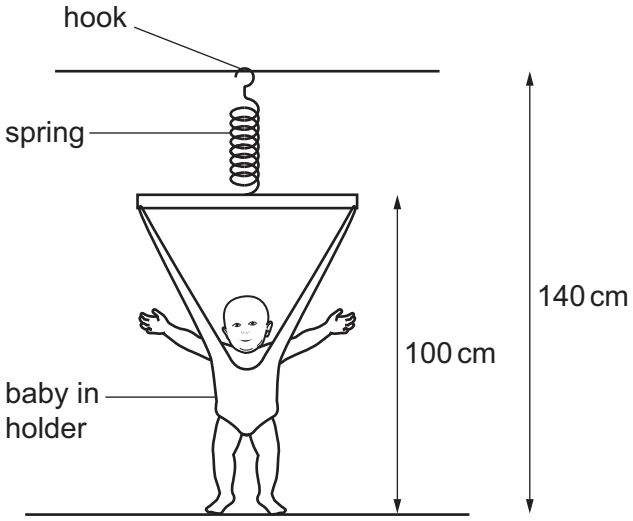

Fig. 2.1

(i) Two springs Q and R are tested to determine their spring constants.

Each spring is tested up to its limit of proportionality.

Define 'limit of proportionality'.

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(ii) Table 2.1 shows the results of the tests.

spring	spring constant / N/cm
Q	7.8
R	1.1

Table 2.1

The total weight of the baby and the holder is 120 N.

Calculate the extension of each spring for this weight.

extension of spring Q = ...

extension of spring R = ...
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(iii) The unstretched length of each spring is 25 cm.

State and explain which spring would be more suitable for the baby bouncer in Fig. 2.1.

spring .....

explanation .....

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